

$$1, E_j = 1) = \alpha_{b0} + \alpha_{b1} + \beta_b > 0).$$

The aggregate supply of human capital to non-encouraged firms satisfies:

$$H_0^S(W_{1:0} | \underbrace{\alpha_b}_{-}, \underbrace{\beta_b}_{-}, \underbrace{\alpha_c}_{+}, \underbrace{\beta_c}_{+}) = \int_{i^*}^1 1di = 1 + \frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_{1:0}), \quad (63)$$

which is strictly decreasing in the relative skill price $W_{1:0}$, strictly increasing in both the common and idiosyncratic productivity benefits of AI chatbots (α_b and β_b), and strictly decreasing in both the common and idiosyncratic adoption costs (α_c and β_c).

Combining Equations (59)-(62), the aggregate supply of human capital to encouraged firms relative to non-encouraged firms satisfies

$$H_{1:0}^S(W_{1:0} | \underbrace{\alpha_b}_{+}, \underbrace{\beta_b}_{+}, \underbrace{\alpha_c}_{-}, \underbrace{\beta_c}_{-}) = \frac{\frac{\exp(\alpha_{b0} + \alpha_{b1})}{\beta_b} \left[\exp(-\frac{\beta_b}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_{1:0})) - 1 \right]}{1 + \frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_{1:0})}, \quad (64)$$

Intuitively, the relative supply of human capital to encouraged firms is higher when the relative skill price is high (inducing more workers to sort into these firms), the productivity benefits of chatbot adoption (α_b and β_b) are large, and adoption costs (α_c and β_c) are low—provided that chatbots are not counterproductive for encouraged adopters.

Skill price differentials. Equilibrium skill price differentials equate the relative supply and demand for human capital at encouraged firms:

$$H_{1:0}^S(W_{1:0}) = H_{1:0}^D(W_{1:0}) \quad (65)$$

Since the supply curve $H_{1:0}^S$ is strictly increasing in $W_{1:0}$ and the demand curve $H_{1:0}^D$ is strictly decreasing, Equation (65) pins down a unique equilibrium skill price differential.

Panel (b) of Figure A.2 illustrates the inverse aggregate supply and demand curves for relative human capital at encouraged firms. The equilibrium skill price differential corresponds to their intersection.

Equilibrium skill price differentials are high when relative demand for human capital, $H_{1:0}^D$, exceeds relative supply, $H_{1:0}^S$. From Equations (58) and (64), this occurs when

encouragement costs are low (F_κ is high), product demand is elastic (ϵ is high), the productivity benefits of chatbot adoption (α_b, β_b) are small, and adoption costs (α_c, β_c) are high:

$$W_{1:0}(\underbrace{F_\kappa}_{+}, \underbrace{\epsilon}_{+}, \underbrace{\alpha_b}_{-}, \underbrace{\beta_b}_{-}, \underbrace{\alpha_c}_{+}, \underbrace{\beta_c}_{+}). \quad (66)$$

We may use the policy rules for encouragement, adoption, and sorting to further bound the equilibrium skill price differentials.

First, in any equilibrium where some but not all firms encourage chatbot use, Equation (19) implies that the relative skill price must lie within:

$$\log W_{1:0} \in [\kappa_{\max}/(1 - \epsilon), \kappa_{\min}/(1 - \epsilon)]. \quad (67)$$

Similarly, in any equilibrium where only some workers sort into encouraged firms, Equation (22) implies that:

$$\log W_{1:0} \in [-\hat{\alpha}_1, 0]. \quad (68)$$

That is, when $\log W_{1:0} > 0$, even non-adopters prefer encouraged firms; when $\log W_{1:0} < -\hat{\alpha}_1$, even adopters prefer non-encouraged firms.

B Data Construction

B.1 Sampling Protocol

Our 2024 survey invited 115,000 workers across 11 occupations. Ideally, we would sample an equal number from each—i.e., 10,450 journalists, 10,450 software developers, etc. However, some occupations in Denmark employ fewer than 10,450 workers. To address this, we follow these steps:

1. If an occupation has fewer than 10,450 workers, we sample all available workers.
2. The remaining invitations are redistributed equally among the other occupations.
3. Workplaces are randomly selected for sampling, and if chosen, all relevant workers (i.e., those in the target occupation) within the workplace are included.
4. Large workplaces can distort the sample balance. To mitigate this, we apply individual-level sampling to the top 2.5% of workplaces (ranked by the number of employees in the relevant occupation), randomly selecting employees using the same sampling probability as in Step 3.
5. To precisely reach our target of 115,000 workers, we make final adjustments by randomly including or excluding workers, independent of their workplace.

The 2023 survey followed a similar protocol, with minor modifications; see Humlum and Vestergaard (2025) for details.

B.2 Survey Sample

Table B.1 outlines how successive sample restrictions define our analysis sample. In total, we obtained about 25,000 complete and valid responses per survey round that can be linked to registry data. The attrition and response rates in our survey are comparable to those obtained in previous Danish surveys (Hvidberg, Kreiner and Stantcheva, 2023). While our main analysis focuses on responses from the 2024 round, we use the 2023 round to examine the dynamics of our estimated effects.

Table B.1: Sample Construction

	<i>2024 Survey</i>		<i>2023 Survey</i>			
	Individuals	Percent of invitees	<i>Main Survey</i>		<i>Follow-Up Only</i>	
			Individuals	Percent of invitees	Individuals	Percent of invitees
1. Invitees	115,000	100.0	100,000	100.0	15,000	100.0
2. Respondents	30,411	26.4	29,067	29.1	4,094	27.3
3. In target occupation(s)	26,925	23.4	25,121	25.1	3,504	23.4
4. Complete responses	25,241	21.9	18,109	18.1	2,561	17.1
5. Linked to registers	24,796	21.6	17,907	17.9	2,559	17.1

Notes: This table outlines how successive sample restrictions define our analysis sample. We conducted two survey rounds in November 2023 and 2024, each inviting 115,000 workers to participate. The 2023 survey included both a main survey and a two-week follow-up, with 15,000 workers invited only to the follow-up. Row 2 reports the number of individuals who responded to the survey. Row 3 shows the respondents who were still employed in one of our 11 target occupations at the time of the surveys. Row 4 presents the respondents who fully completed the survey questionnaire. Row 5 indicates the complete responses that could be linked to registry data.